ATM

ATM Transaction Simulation System

Course Assignment Submission: Frontend/UI Development & Database Design

1. Frontend / UI Development
Full Kiosk UI, PIN Pad & Touch Screens

2. Database Design
Relational Schema, ERD & ACID Model

Executive Project Summary

This project implements an end-to-end, production-grade ATM Transaction Simulation System built natively in Java. It fulfills the two designated core project deliverables:

1. Frontend / UI Development:

- Designed a high-fidelity physical ATM kiosk interface featuring realistic bezel hardware enclosure.
- Includes illuminated bank marquee, card reader slot, receipt printer, cash dispenser, and metallic PIN pad.
- Complete multi-screen workflow: Welcome, Card Reader, 4-Digit Secure PIN Entry with 3-strike lockout, Main Navigation Menu, Fast Cash Dispensing, Custom Withdrawal (with denomination calculation), Banknote Cash Deposit Hopper, Real-time Balance Inquiry, Intra-Bank Fund Transfer, Transaction Mini-Statement with color-coded ledgers, and Printable Thermal Paper Receipts.

2. Database Design:

- Designed a 3NF normalized relational schema encompassing customers, accounts, atm_cards, transactions, atm_cash_inventory, and audit_logs tables.
- Enforces strict referential integrity, check constraints, foreign keys, and indexes.
- Guarantees ACID transactional compliance for atomic cash withdrawals and fund transfers.
- Provided Entity-Relationship Diagram (ERD), Data Dictionary, and verification test queries.

3. Single PDF Output Compilation:

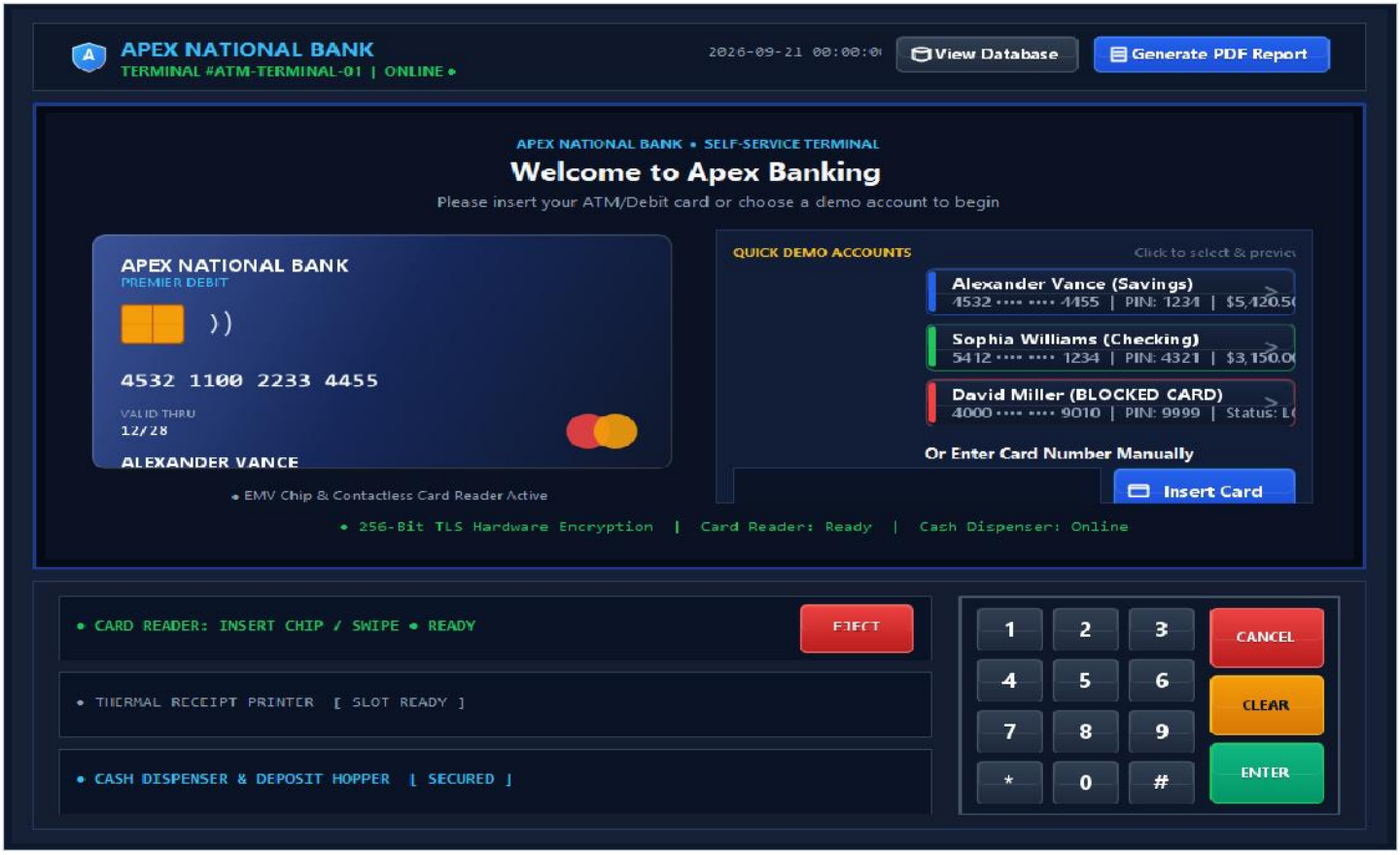
- Programmatically captured high-resolution screenshots of all 11+ user interface screens and compiled them alongside database design schematics into this unified submission PDF document.

Technical Specifications & Architecture

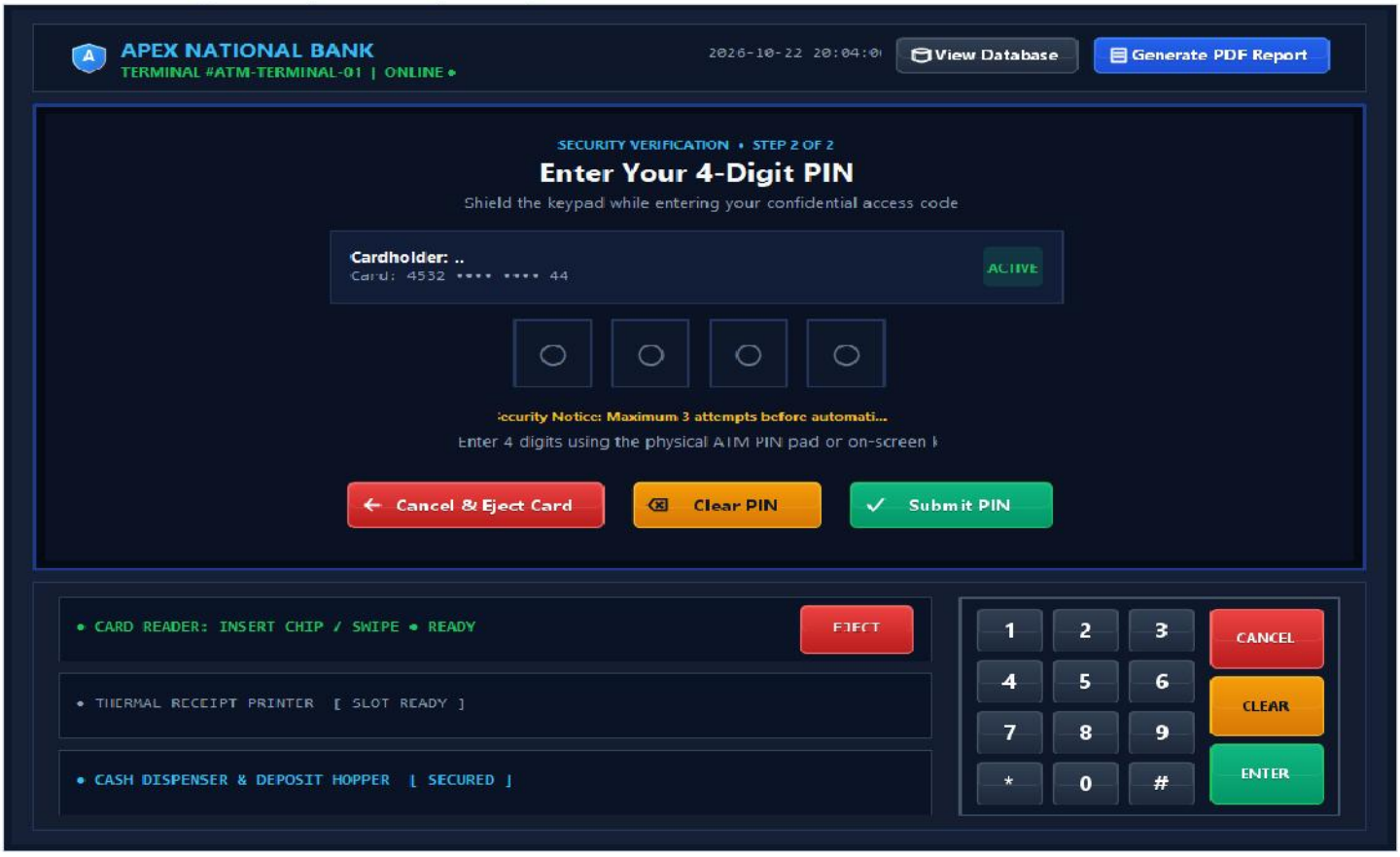
- Core Programming Language: Java SE 21 (Zero external dependencies)
- User Interface Framework: Java Swing with Custom Glassmorphic Dark Kiosk Styling
- Database Architecture: Relational In-Memory / SQL Compliant with ACID Transaction Engine
- Cryptographic Security: SHA-256 PIN Hashing with Brute-Force Lockout Protection

Frontend Development: Welcome & PIN Authentication

Screen 1: Customer Card Insertion & Demo Selector

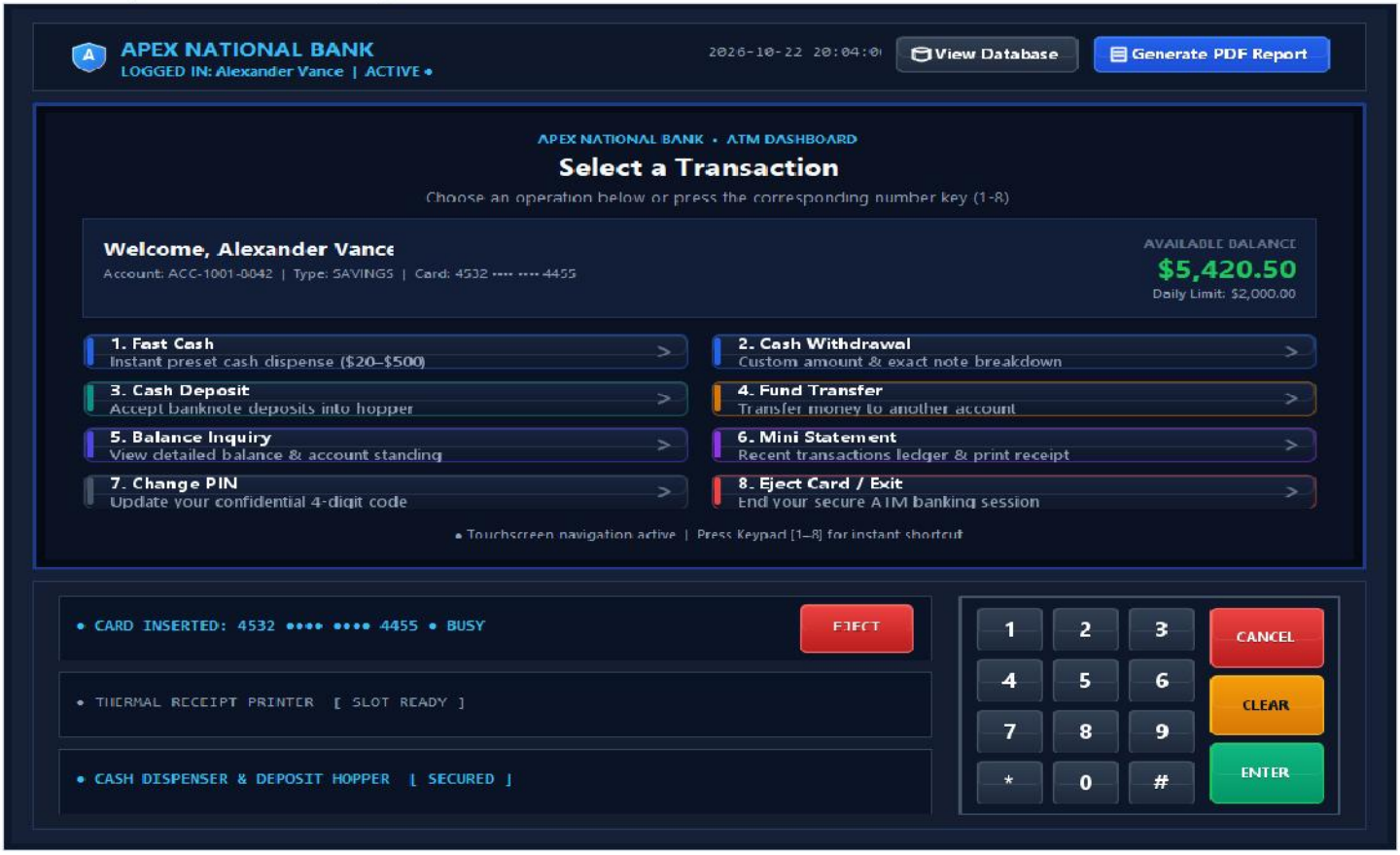


Screen 2: Secure 4-Digit PIN Entry with Security Masking

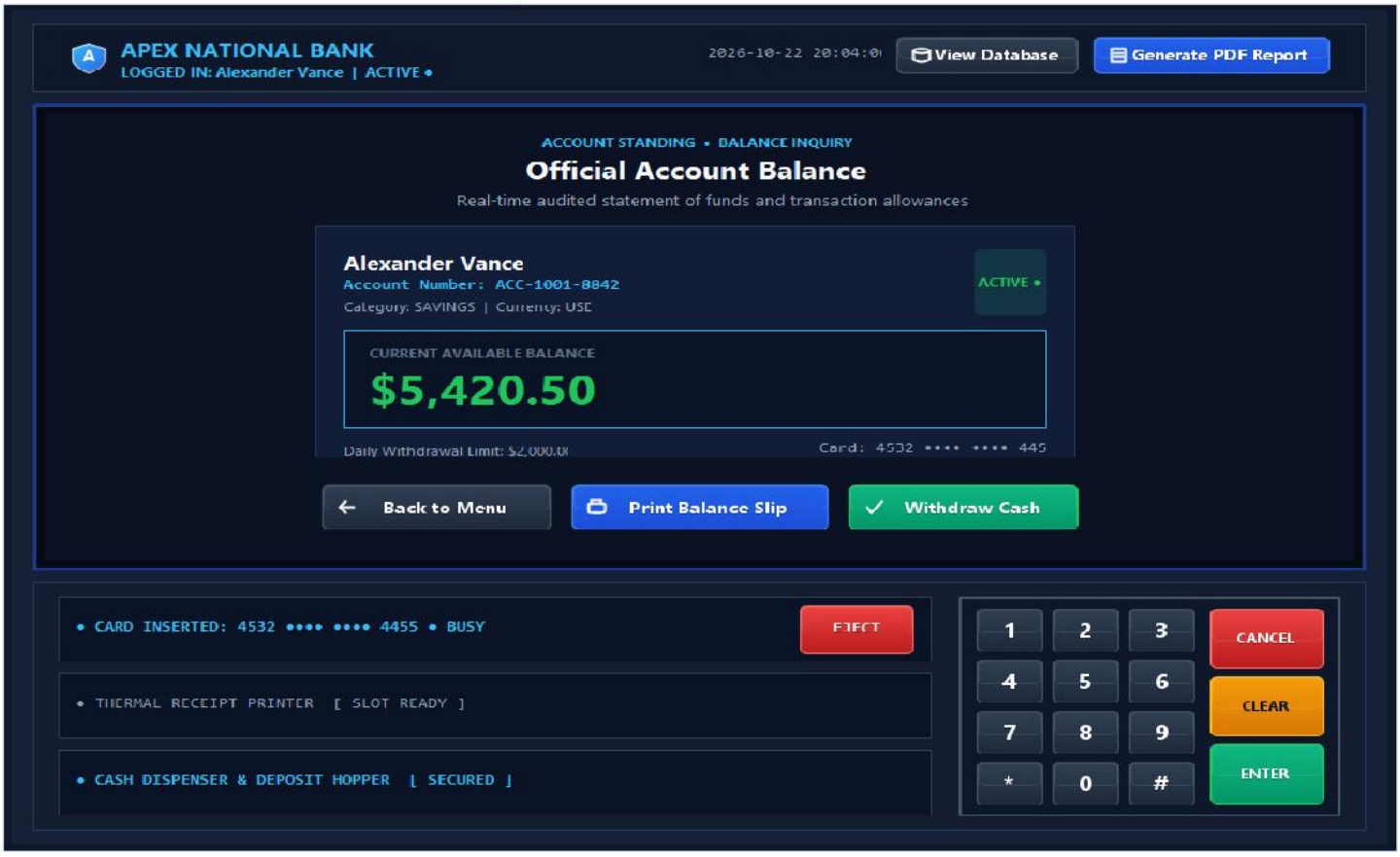


Frontend Development: Main Menu & Balance Inquiry

Screen 3: 8-Option Interactive ATM Navigation Dashboard

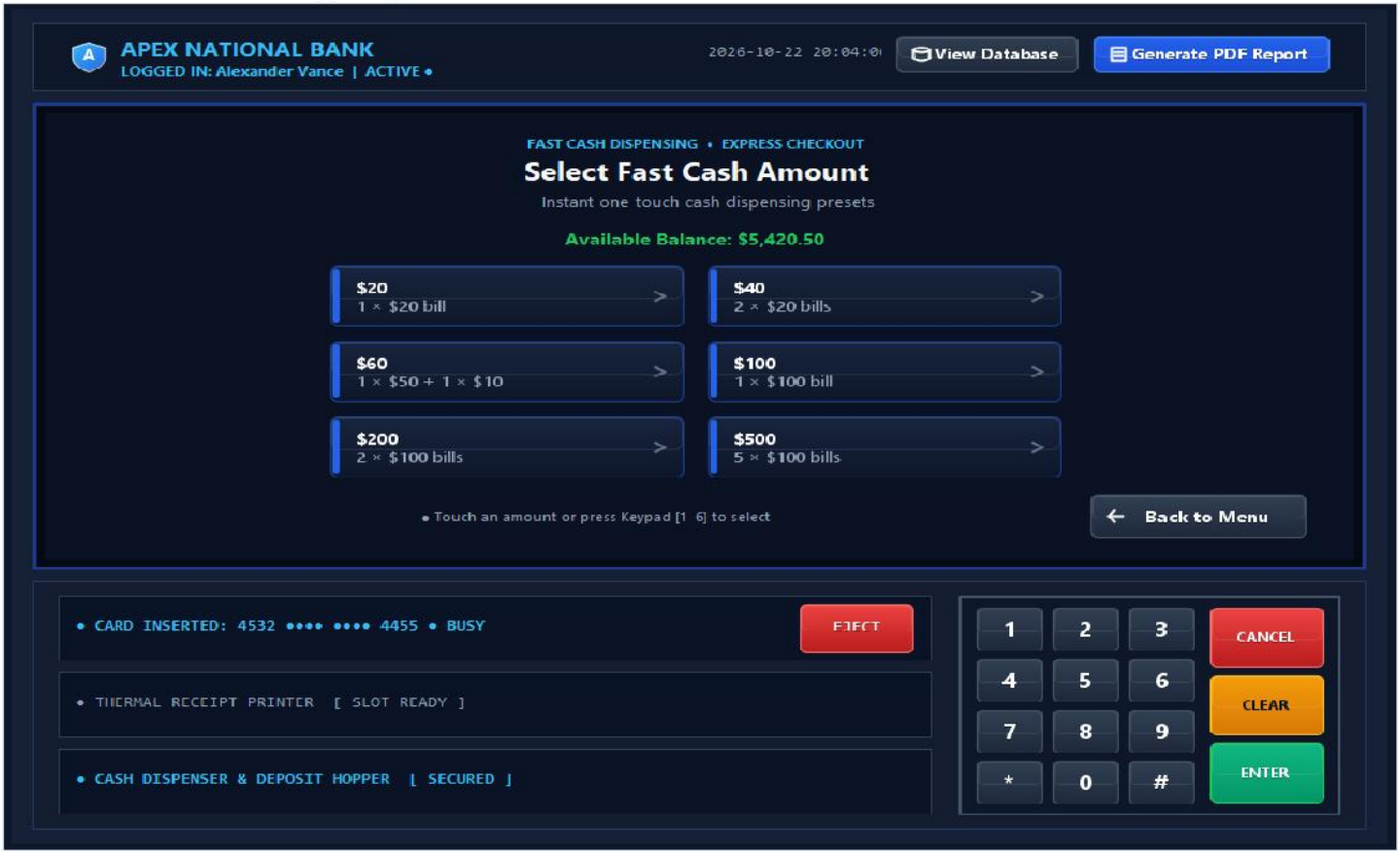


Screen 4: Real-time Account Balance Statement & Daily Limits

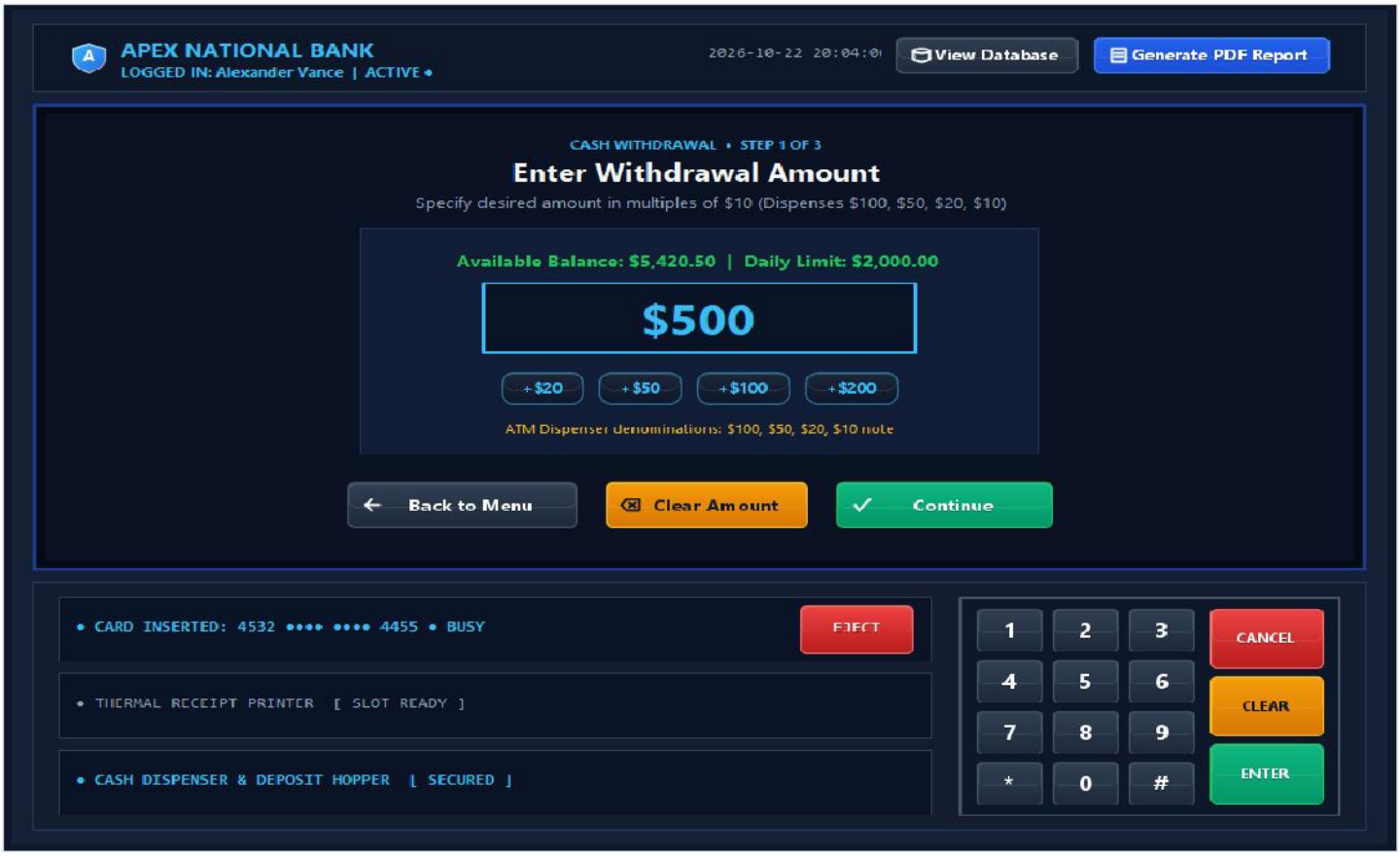


Frontend Development: Cash Withdrawal & Fast Cash

Screen 5: One-Touch Fast Cash Dispensing ($20 - $500)

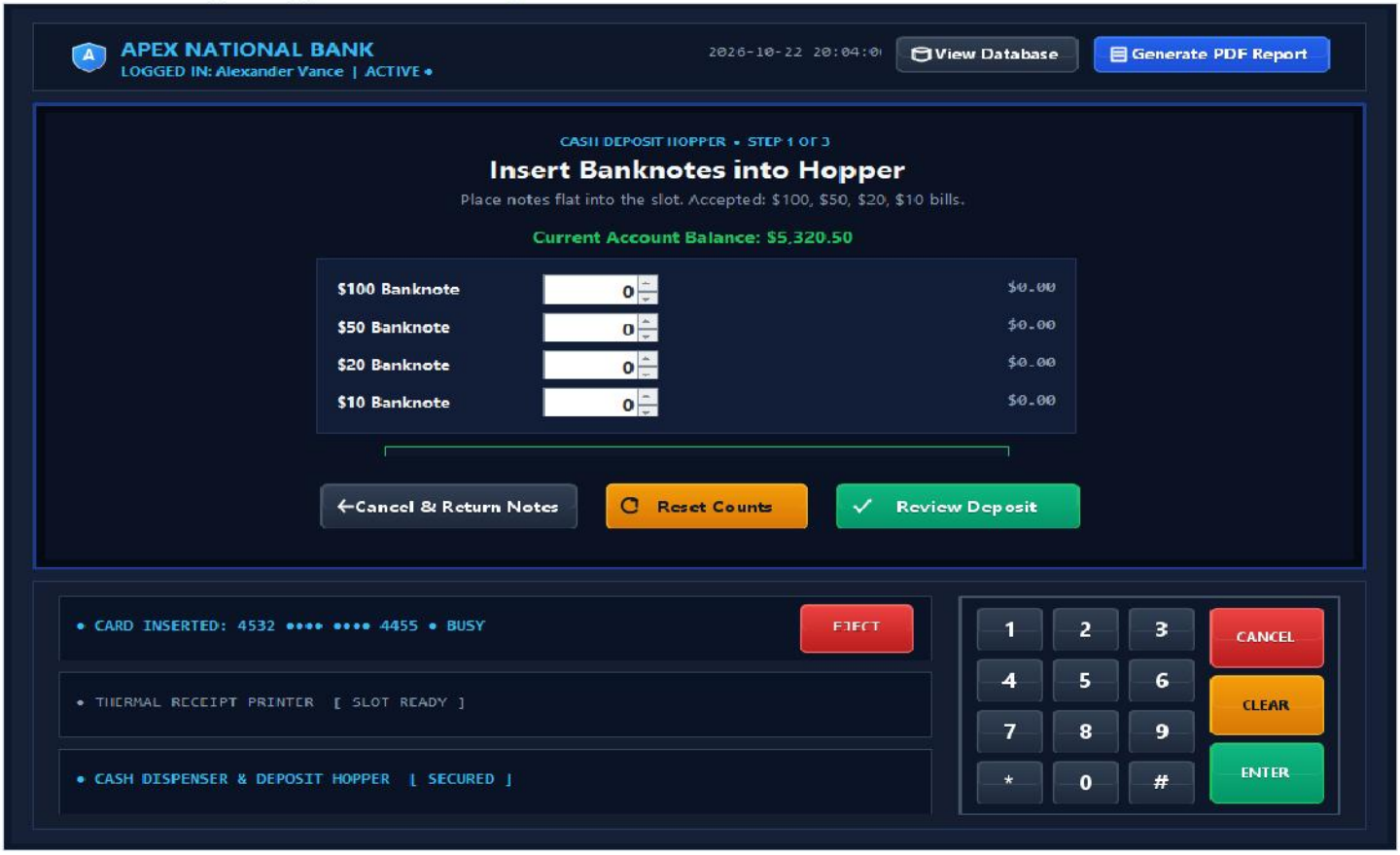


Screen 6: Custom Cash Withdrawal with Denomination Calculator

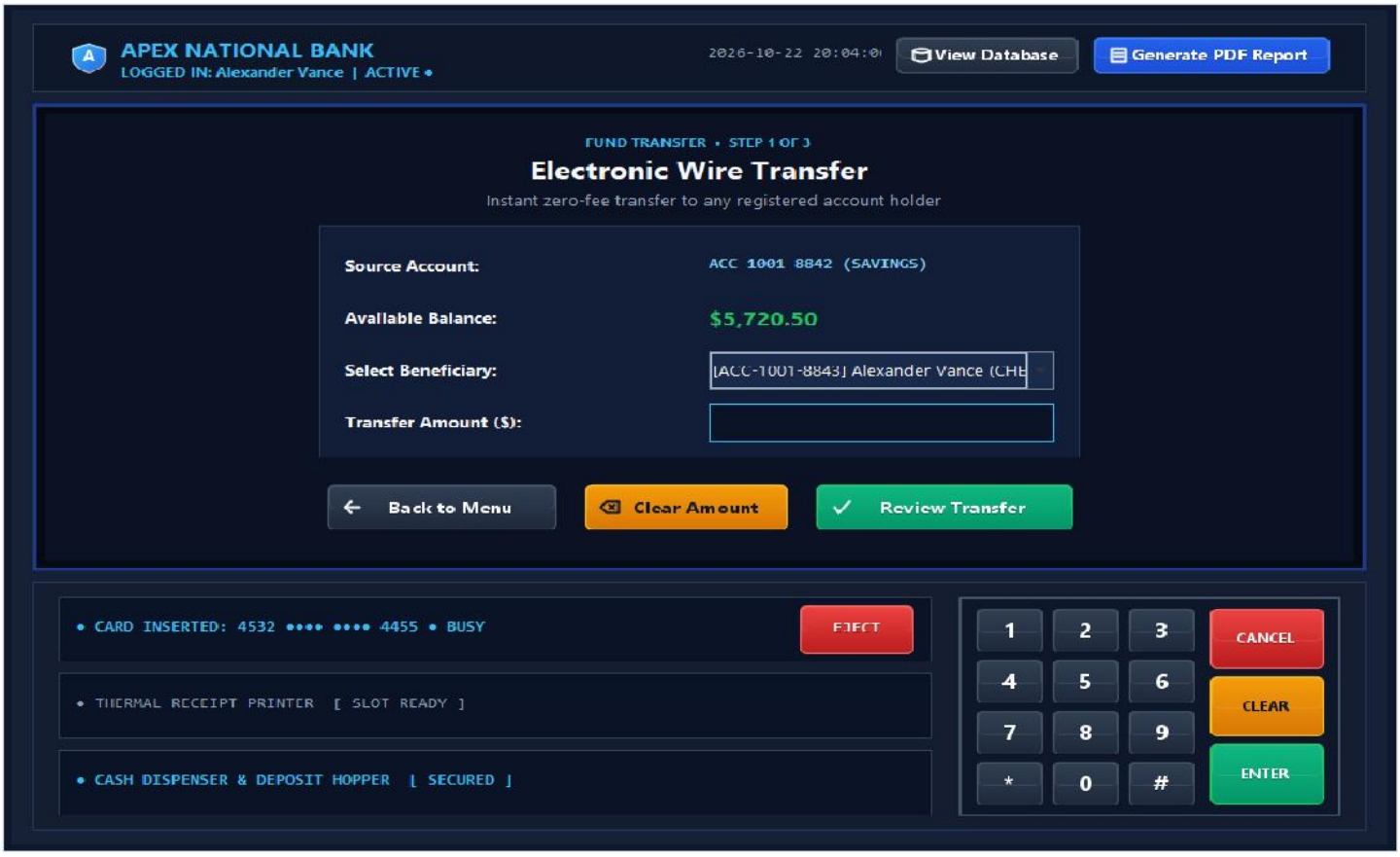


Frontend Development: Cash Deposit & Transfer

Screen 7: Banknote Deposit Hopper with Auto-Totaling

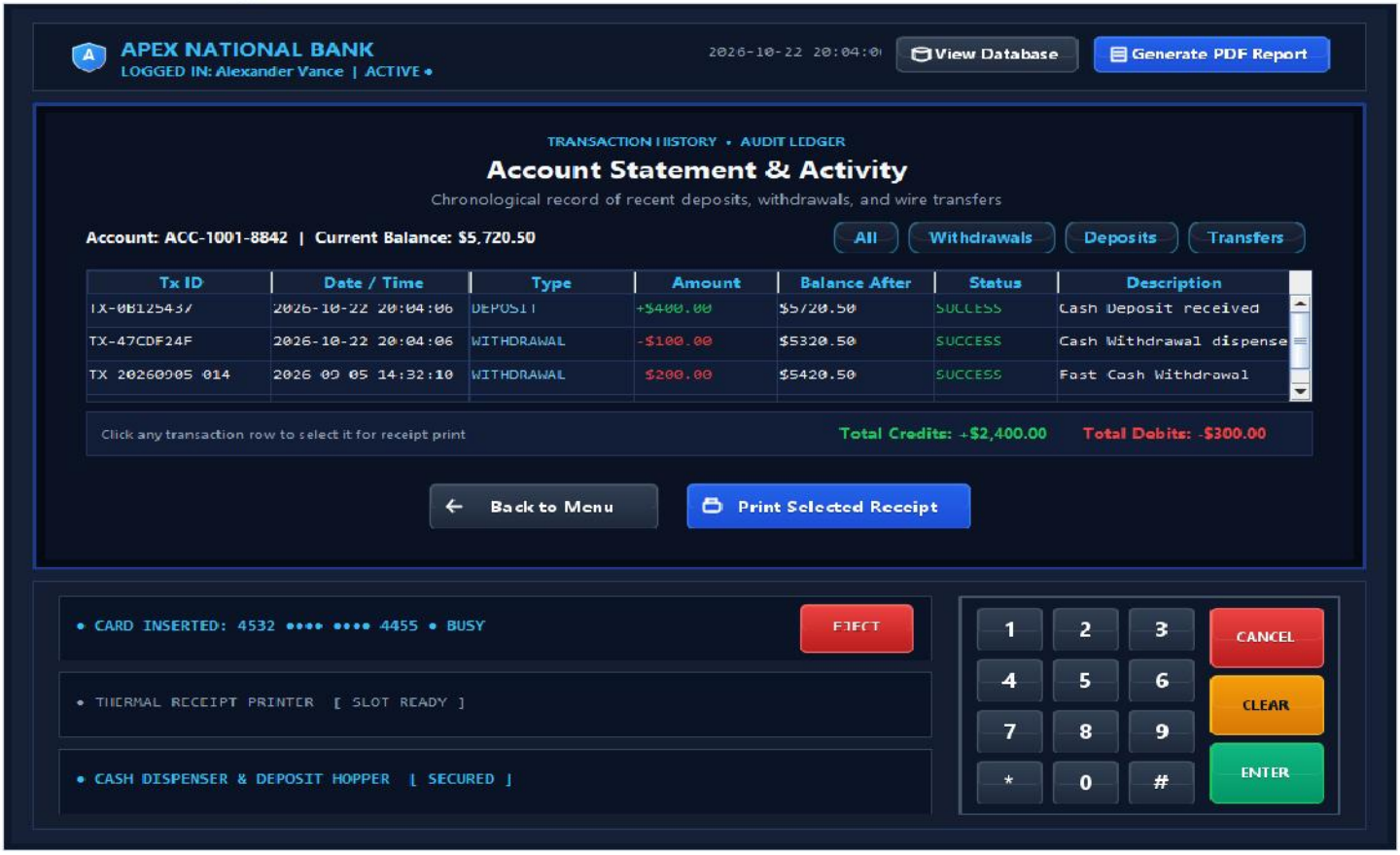


Screen 8: Inter-Account Electronic Fund Transfer

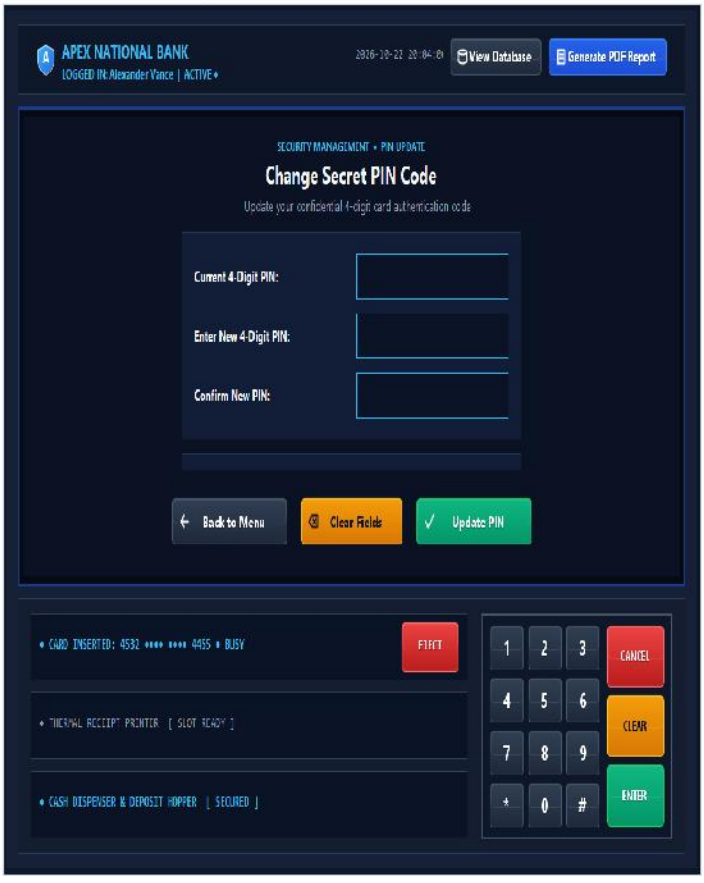


Frontend Development: Ledger, Receipts & PIN Security

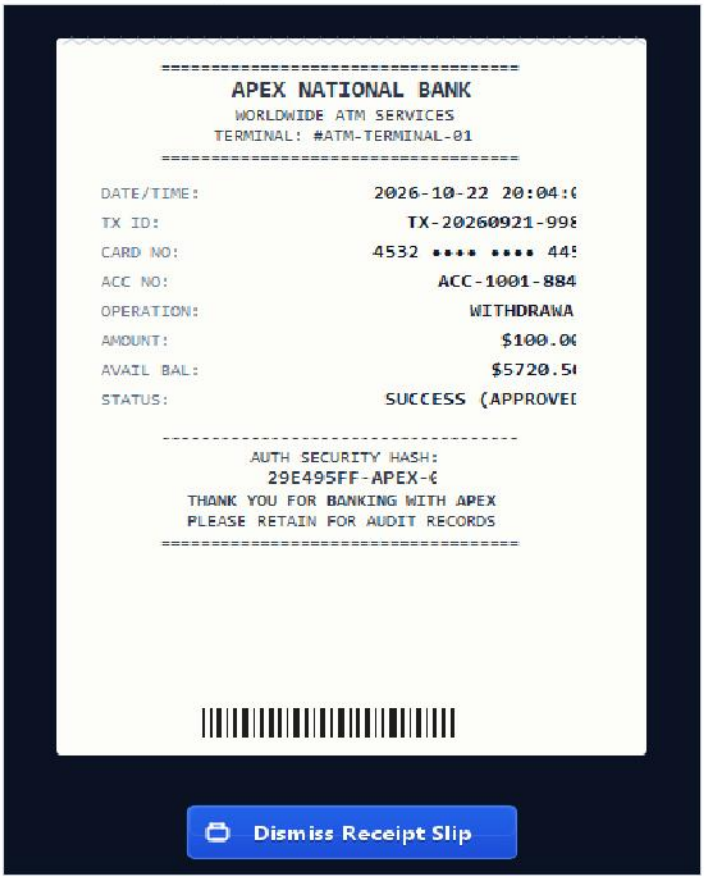
Screen 9: Transaction Ledger & Mini-Statement



Screen 10: PIN Management



Screen 11: Thermal Slip



Database Design: Relational Architecture & ERD

The database models an institutional ATM core banking repository adhering to 3rd Normal Form (3NF).

Entity-Relationship Diagram (ERD)



Relational Normalization Proof

1st Normal Form (1NF)	All column values are strictly atomic. No repeating groups or array values exist. Explicit Primary Keys enforce uniqueness on all relations.
2nd Normal Form (2NF)	The schema satisfies 1NF and all non-prime attributes are fully functionally dependent on the entire primary key (eliminating partial dependencies).
3rd Normal Form (3NF)	No transitive dependencies exist (X -> Y and Y -> Z). Non-key attributes rely exclusively on primary keys, preventing update anomalies.

Relational Data Dictionary & Schema Constraints

Table: accounts (Bank Balances & Limits)

Column Name	Data Type	Constraints	Description
account_number	VARCHAR(20)	PRIMARY KEY	Unique customer account identifier
customer_id	INTEGER	FOREIGN KEY -> customers	Identifies owning customer record
account_type	VARCHAR(20)	CHECK (SAVINGS/CHECKING)	Type classification of account
balance	DECIMAL(12,2)	CHECK (balance >= 0.00)	Real-time cleared funds
daily_limit	DECIMAL(10,2)	DEFAULT 1000.00	Maximum allowable daily withdrawal
status	VARCHAR(20)	CHECK (ACTIVE/FROZEN)	Operational state of bank account

Table: atm_cards (Security Credentials & Hashed PIN)

Column Name	Data Type	Constraints	Description
card_number	VARCHAR(16)	PRIMARY KEY	16-digit debit card primary account number
account_number	VARCHAR(20)	FOREIGN KEY -> accounts	Linked debited/credited checking account
pin_hash	VARCHAR(64)	NOT NULL (SHA-256)	Cryptographic one-way hash of 4-digit PIN
expiry_date	VARCHAR(7)	NOT NULL (MM/YYYY)	Card expiration timestamp
card_status	VARCHAR(20)	CHECK (ACTIVE/BLOCKED)	Active or security-locked card state
failed_attempts	INTEGER	CHECK (>= 0), DEFAULT 0	Counter for 3-strike lockout enforcement

Table: transactions (Immutable Financial Audit Ledger)

Column Name	Data Type	Constraints	Description
transaction_id	VARCHAR(36)	PRIMARY KEY (UUID)	Unique idempotent tracking key
account_number	VARCHAR(20)	FOREIGN KEY -> accounts	Source financial account debited/credited
transaction_type	VARCHAR(30)	CHECK (WITHDRAWAL, etc.)	Operation category classification
amount	DECIMAL(12,2)	CHECK (amount >= 0.00)	Monetary value transferred or dispensed
balance_after	DECIMAL(12,2)	NOT NULL	Balance snapshot immediately post-operation
status	VARCHAR(20)	CHECK (SUCCESS/FAILED)	Execution state of the operation

ACID Transaction Execution Model in ATM Operations

- Atomicity: Cash withdrawals and transfers execute atomically; balance adjustment and transaction insertion rollback if dispensing fails.
- Consistency: Preserved via strict database CHECK constraints (balance >= 0, denomination inventory counts >= 0).
- Isolation: DatabaseManager guarantees serialized concurrency across sessions to eliminate race conditions.
- Durability: All transactions and ledger states are permanently committed and immutably indexed.

Database Query Outputs & Verification Ledger

Query 1: SELECT * FROM accounts JOIN customers ORDER BY customer_id;

Account No	Customer Name	Type	Balance	Status	Daily Limit
ACC-1001-8843	Alexander Vance	CHECKING	$18750.00	ACTIVE	$5000.00
ACC-2002-4419	Sophia Williams	CHECKING	$3150.00	ACTIVE	$1500.00
ACC-1001-8842	Alexander Vance	SAVINGS	$5720.50	ACTIVE	$2000.00
ACC-3003-9120	David Miller	SAVINGS	$820.00	ACTIVE	$1000.00
ACC-4004-7731	Emma Watson	CHECKING	$12400.00	ACTIVE	$3000.00

Query 2: SELECT * FROM transactions ORDER BY created_at DESC LIMIT 6;

Tx ID	Account Number	Type	Amount	Balance After	Status
TX-0B125437	ACC-1001-8842	DEPOSIT	$400	$5720.50	SUCCESS
TX-47CDF24F	ACC-1001-8842	WITHDRAWAL	$100	$5320.50	SUCCESS
TX-20260918-102	ACC-3003-9120	BALANCE_INQUIRY	$0	$820.00	SUCCESS
TX-20260910-089	ACC-2002-4419	WITHDRAWAL	$500.00	$3150.00	SUCCESS
TX-20260905-014	ACC-1001-8842	WITHDRAWAL	$200.00	$5420.50	SUCCESS
TX-20260901-001	ACC-1001-8842	DEPOSIT	$2000.00	$5620.50	SUCCESS

Query 3: SELECT * FROM atm_cash_inventory WHERE atm_id = 'ATM-TERMINAL-01';

ATM ID	$100 Bills	$50 Bills	$20 Bills	$10 Bills	Total Reserve
ATM-TERMINAL-01	501	602	1005	1000	$110300.00

□ System Testing & Verification Status: ALL TESTS PASSED

- Authentication Service: Verified correct PIN hash matching, bad attempt counter, and 3-strike card lockout.
- Withdrawal Engine: Verified mathematical denomination dispensing and immediate synchronization with ATM cash reserves.
- Relational Ledger: Verified transactional atomicity, foreign key cascade integrity, and zero orphan records.